

VIPUL RAJ JHA

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EDUCATION

Indraprastha Institute of Information Technology, Delhi (IIITD)

Bachelor of Technology - *Computer Science and Social Sciences (Dean's List Awardee)*

School of Excellence, Kalkaji, New Delhi(PCMB, Grade 12)(Mar 2019 - June 2020)

Panchsheel Public School, New Delhi (Grade 10) (Mar 2017 - June 2018)

Aug 2021 - May 2025
(ongoing)

CBSE: 89.80/100

CBSE: 90.40/100

TECHNICAL SKILLS

Languages C++, C, Java, Python, JavaScript

Backend Tools MySQL, React.js, C#

Tools and Technologies Git, Docker, Kubernetes, IntelliJ IDEA, NumPy, Pandas, Unity, Computer Networks

Cloud AWS

Operating Systems Windows, LINUX, Arch Linux

Areas of Expertise Data Structures, Algorithms, OOPs, DBMS, MySQL, Problem Solving, High Frequency Trading(HFT)

Soft Skills Proactive, High Aptitude, Effective Communicator, Collaborative

EXPERIENCE

ETiDM Lab (VR Developer)

Jan 2024 - Aug 2024

- Designed and implemented 3D modeling software in Unity with C#, HTML, CSS and JavaScript taking inspiration from Unity and Blender to facilitate the **creation and manipulation of 3D shapes with a 50% increase in user interaction speed.**
- Formulated immersive tools for enhanced 3D visualization and manipulation, **supporting adjustments across 3 axes and improving user experience by 40%.** [Link](#) , [Demo](#), [Work Recommendation](#)

HCD Lab (Research Intern between IIITD & Ariel Uni. ,Israel)

July 2023 - May 2024

- Analyzed design methods' impact on ideation and cost, enhancing collaborative brainstorming on climate-resilient building designs and diverse subjects using optimized Java modules like Linkoder, **boosting research efficiency by 50%.**[Publication](#)

PROJECTS

CourseBot360 [source code]

Jan 2024 – April 2024

HTML, CSS, JavaScript, OpenAI API

- Designed and developed a chatbot to assist students with course and elective selection, incorporating robust backend data handling and an intuitive, user-friendly interface.
- Achieved a **6x improvement in course search efficiency, reducing manual queries by 70%** and significantly enhancing the student experience with personalized recommendations.

Web Scraping Tool for Research Paper Abstracts

Sept 2024 - Present

Python, BeautifulSoup, Selenium, PubMed API, Pandas, NumPy

- Developed a Python web scraping tool to **extract over 10,000 abstracts from Google Scholar and PubMed**, enabling efficient exploration of ML/DL models and user-specific queries.
- Utilized a large language model to process and analyze data, **improving research efficiency in drug discovery and material science by 40%.**

Cab Booking System [source code]

Jan 2023 - April 2023

MySQL, Javascript, PHP, Pandas, HTML, CSS (Polymorphism, Inheritance, Interfaces)

- Developed admin functionalities to view detailed data for **100+ users, 10+ cabs, 100+ rides, and 50+ drivers.**
- Implemented user features for login/sign up, balance management, and payment processing, **supporting over 1,000 of them using SQL for efficient database operations.**

Tank Stars [source code]

Oct 2022 - Dec 2022

Java, libGDX, Object Oriented Programming

- Engineered a robust clone of the popular game Tank Stars using the LibGDX framework in Java, applying OOP principles and design patterns (**Singleton, Factory, Observer**), with physics-based simulations, customizable tank mechanics, and **real-time 2-player multiplayer functionality** for an immersive experience.

Awards/Achievement

Achieved **Dean's List for Academics** (2023-2024), **Specialist** at [Codeforces](#) (**1485**), Top 4% on [LeetCode](#)(total problem solved), **World Rank 979**, in Codeforces Round 994 out of 35k+ candidates, cleared **JEE Advanced**(2021), **City finalist** in National Aptitude Test(NIE, TOI, 2017), **Top 4% in Young Turks** by Naukri.com, [Ethical Hacking Certification](#), and **taught 100+ underprivileged students** for JEE and NEET. [Link](#)